DISGAS

DISGAS (passive DISpersion of GASes and particles) is a Eulerian model for passive dispersion of diluted gas and fine dust particles.

References:

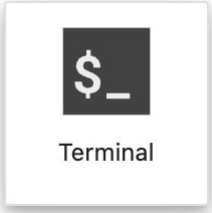
A. Costa, G. Macedonio, Chiodini G., 2005. Numerical model of gas dispersion emitted from volcanic sources. Annals of Geophysics, Vol. 48: 805-815. https://www.annalsofgeophysics.eu/index.php/annals/article/view/3236

Granieri D., Costa A., Macedonio G., Chiodini G., Bisson M. (2013) Carbon dioxide in the city of Naples: contribution and effects of the volcanic source, J. Volcanol. Geotherm. Res., Vol. 260: 52-61, doi: 10.1016/j.jvolgeores.2013.05.003 https://www.sciencedirect.com/science/article/pii/S0377027313001443

Costa A., Macedonio G. (2016) DISGAS: A model for passive DISpersion of GAS, Rapporti tecnici INGV, N. 332, Istituto Nazionale Di Geofisica e Vulcanologia, Italy http://datasim.ov.ingv.it/download/disgas/manual-disgas-2.0.pdf

Activity: DISGAS on VICTOR

- Login to VICTOR and select a small server size.
- In The Launcher, open the terminal using the icon:

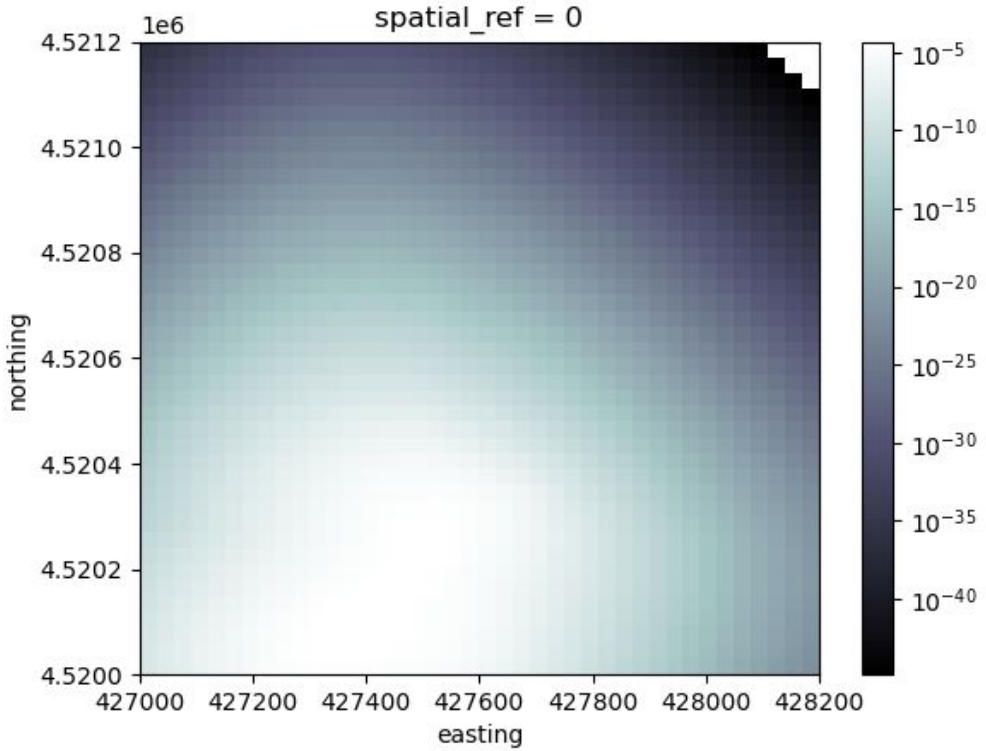


- Type *victor setup* and press enter.
- Select DISGAS from the list that appears by typing its corresponding number (3) and pressing enter.
- In the home directory (on the left of your screen) the  folder will load. Double click on the folder to open it.

- Within the Disgas folder, you will see the Notebook  Open it by double clicking on it.

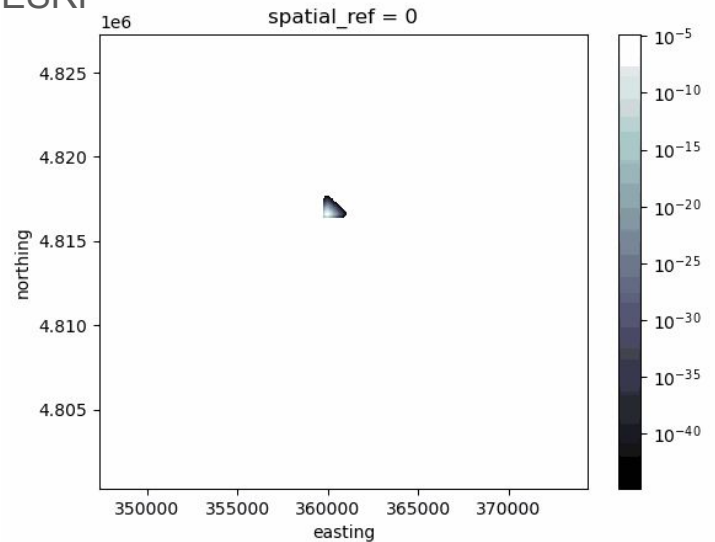
- Run the model by clicking the cells in the notebook: 

Expected Output:



Now let's run it for the ESRP

- Copy all the files in `shared/Education/2025 Idaho Workshop/DISGAS` folder to your own Disgas folder.
- The changes we made to the files are:
  - We used QGIS to create a `.grd` DEM file for a segment of the ESRP
  - `source.dat` now include a point source located in the ESRP
  - `winds.dat` now has a station location in the ESRP
- Open the notebook: ESRP_disgas.ipynb
- Run all the cells like before
- The result should like like this:



Putting the result in context

- In the Launcher, click the D icon to launch a virtual graphical desktop
- Double-click on the Q icon to launch QGIS
- Start a new project
- Add the Google Earth layer: Layer→ Add xyz layer → Google Earth
- Add the model output layer: Layer → Add raster layer and select a file from the output files folder inside your Disgas folder
- Set the projection of the output layer to UTM 12N
- Set the zero-value cells to transparent: Double-click the layer, under the Transparency tab, write 0 in the “other NoData values” line
- Set the colors of the layer to your preference (under “Symbology”, change RenderType to “Single band pseudocolor” and choose a color scheme).

Expected result:

